

AMBIENT SYSTEMS

Climate At Your Service

Investment Deck Apr 2021

The Future of Farming is Indoors



By 2050, the world is predicted to reach

9.7B

in population and to feed everyone would require increase in food output by

70%

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Traditional Farming Methods are increasingly unreliable



Global Warming has brought about issues such as

Rising Temperatures, Irregular Climates

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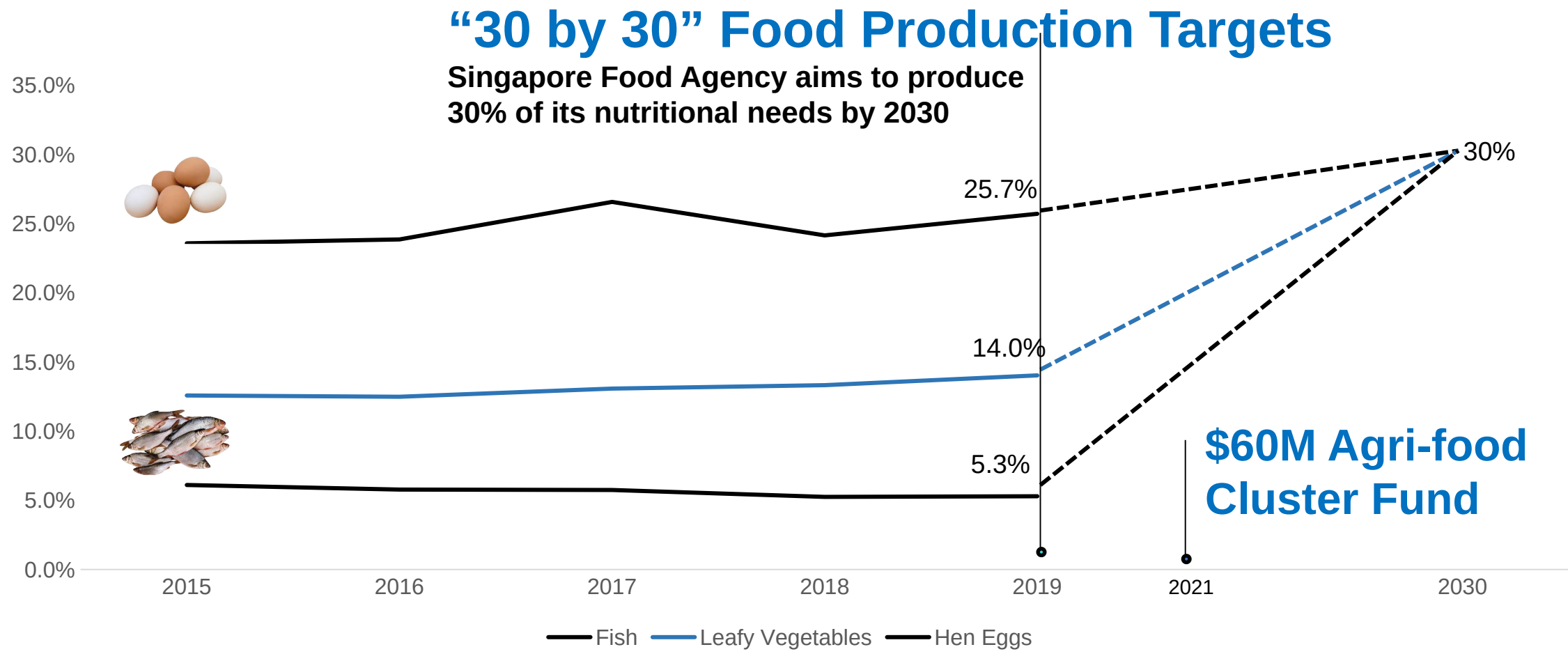
Covid-19 has disrupted traditional farming



Traditional Farms were hit hard by the Covid-19 Pandemic

Quarantines, Disruption to Supply Chains

Local farms sees supports from local government sectors



*Agriculture sector of Singapore is engaged mainly in the production of eggs, fish and vegetables for local consumptions
(1) Fish only includes seafood from coastal and land-based farm production and excluding local landings of seafood (seafood catches from local waters)
Source: Agri-food and Veterinary Authority, Singapore Food Agency, National Parks Board

Bringing the optimal environment conditions indoors

OPTIMAL CONDITIONS DEPENDS ON SEVERAL FACTORS

Day-time Temp: 20-24C
Night-time Temp: 10-12C
Humidity: Low humidity
(Crown): 18C
Light Intensity: 800-1200 micro-mol/m2/s PPF



FARMERS HAVE NO CONTROL ON BUILDING CONDITIONS

No access to building data

QUALITY CROP ARE TIME-SENSITIVE

Crop growth requires real-time monitoring

LACK OF DYNAMIC SOLUTION FOR OPTIMAL ENVIRONMENT

No infrastructure integration with Air Conditioning and Machine Ventilation (ACMV)

Agri-tech remains key area of investment in Southeast Asia

Farm Management Software, Sensing & IOT is the top 5 most invested Agri-tech category in the region (2019)



$423M

Invested

+33%

YoY Investment Growth

+41%

YoY Deal Growth

Source: 2020 , Agfunder

Tam, Sam, Som

[In Progress]

Ambient Platform: Climate-At-Your-Service



Connected.
Centralized.
Automated.

Edited

Video Demonstration: Ambient MVP

Highlights:

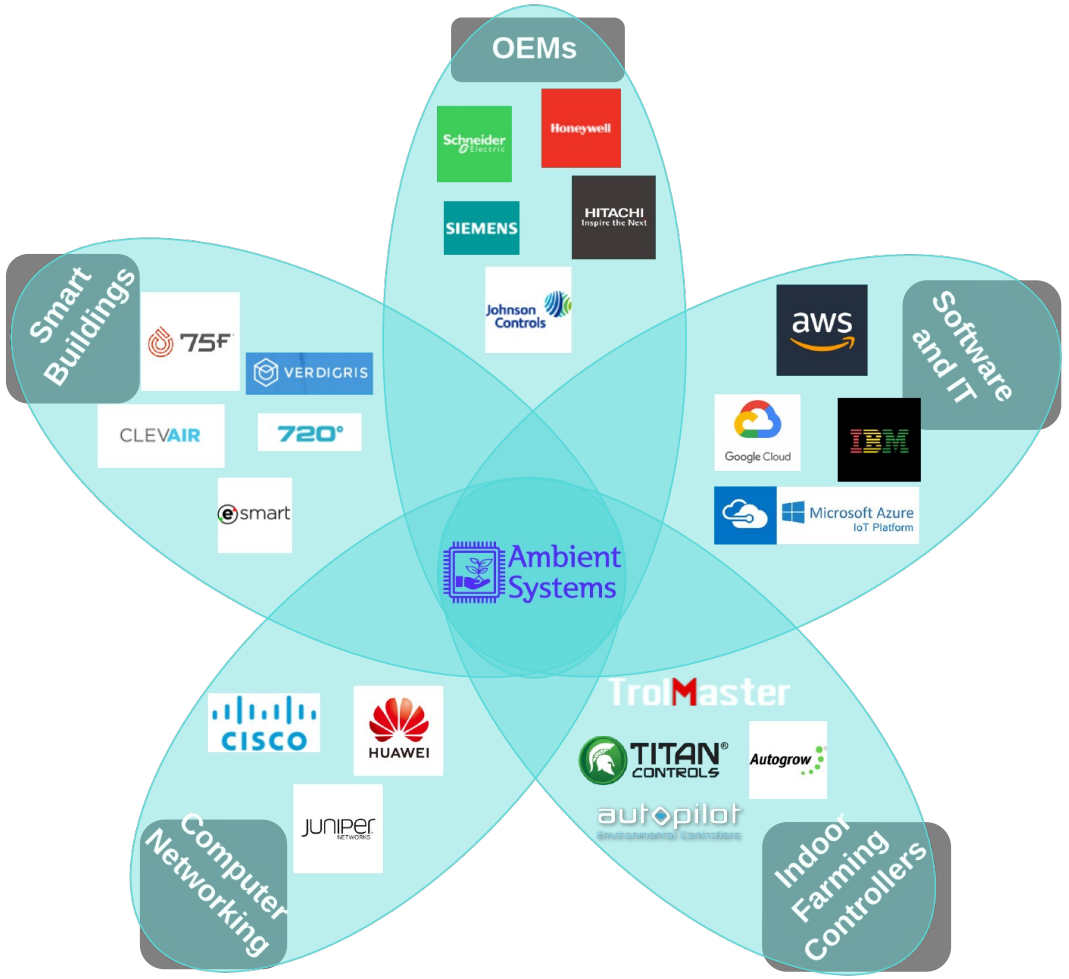
- installation of the hardware at pilot site
- software dashboard
- example of the model --> actual data being analyzed and translated into actionable inputs

[In Progress]

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Edited

Consolidating Current Urban Farming Technologies into a Turn-Key Solution

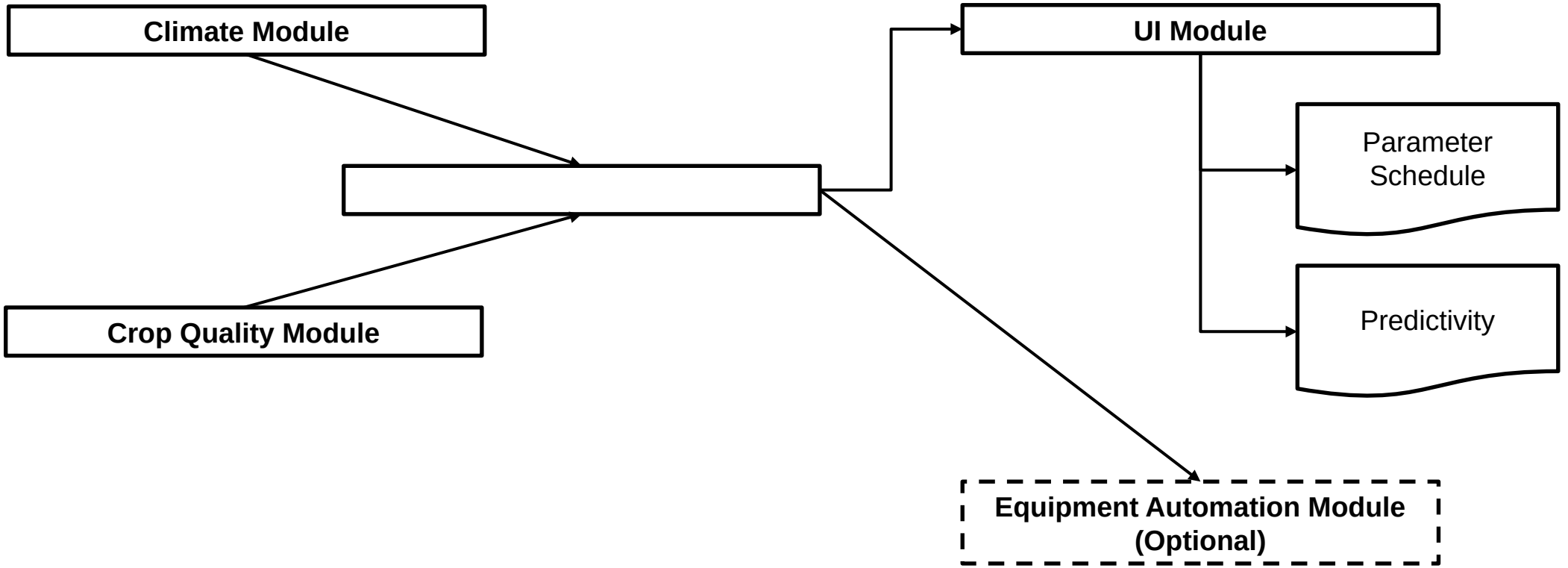


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Ambient Technology (Hardware-software Co-design)

Edited

Ambient's secret sauce: Machine Learning Optimization Model



Patent focuses on climate optimization specific to indoor farming

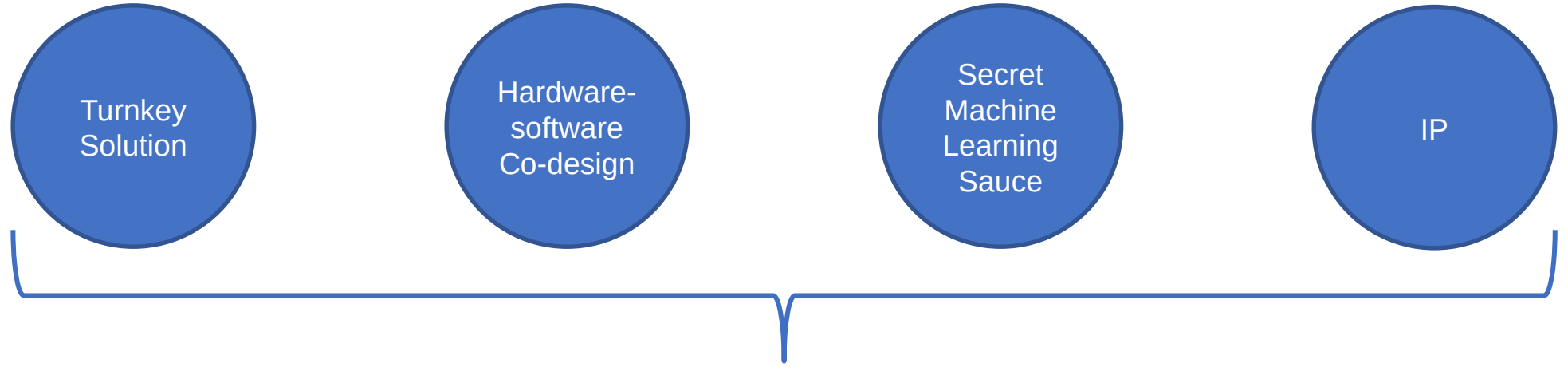
Power of Data~

How do you gain from predictability? Apart from cost savings and improved yield

Show SinGrow Dashboard

[In Progress]

Technology-enabled competitive advantage



Multiple Value Streams

Cost Reduction	**Revenue Generation**
Utility savings (energy efficiency)	Increase Yield
Reduction in manhours for highly trained engineers	Reduce Crop Loss (from sickness)
Lower insurance premiums (thanks to data access)	Enhanced Harvest Prediction

Value-based Pricing driven by our

For Urban Farms

l Pricing Table for 1 Small Sized Farm
(220sqft or 20.4m2)

[In Progress]

SGD($)			
Implementation Cost	Hardware	**Quotation**	IoT Sensors Device & Installation Integration
	Installation	**Quotation**	**One-time cost**
Operational Cost	Monthly Data Management	**$1,000.00**	**Recurring cost**
	Monthly System Maintenance	**$500.00**	Higher Crop Yield Higher Quality Harvest Time Prediction
Use case specific	Monthly Estimated (% of Crop Value)	**5%**	

How do you gain from predictability? Apart from cost savings and improved yield

Our Financial Projections

Base	Y1	Y2	Y3	Y4	Y5
Number of farms managed	14	45	141	354	566
Area of Farm Size Managed (m2)	1,164	3653	11,466	28,873	46,226
No. of Container Farms Equivalent	57	179	562	1415	2266

	Bear	Base	Bull
Runway (Months)	20	17	15
Funding Gap (S$)	$510K	**$455K**	$415K

Software Subscriptions (SGD$)

Revenue	$302,069	$1,004,417	$3,351,733	$9,768,693	$20,031,648
Gross Margin	91.15%	91.65%	92.14%	92.63%	93.10%
P&L	($402,662)	$94,621	$1,910,903	$7,095,040	$15,262,342
	($402,662)	($308,041)	$1,602,862	$8,697,902	$23,960,243
Discount Rate	10%				
Net Present Value	**$21,401,829**				

What will our fund be used for and how it will help in the next milestone

Ambient's Climate As A Service solution can be seamlessly deployed in any industries that require a controlled environment









List Potential LOIs

[In Progress]

About Us

Ambient is a deep-tech spin-off from the National University of Singapore that is developing an integrated hardware-software solution for climate management of vertical farms.

The product leverages intellectual property on securely combining various sources of data and machine learning to enhance building performance.

Product Development

- **MComp, Computer Science**
  National University of Singapore
- **BASc, Civil Engineering, Sustainable Energy**
  University of Toronto
- **Serial Entrepreneur**
  5 years of experience in cleantech & deep tech start-ups (YC-backed)

Corporate Development

- **BBM, Finance**
  Singapore Management University
- **Strategy and Fundraising**
  3 years of experience across e-commerce, gaming and payments



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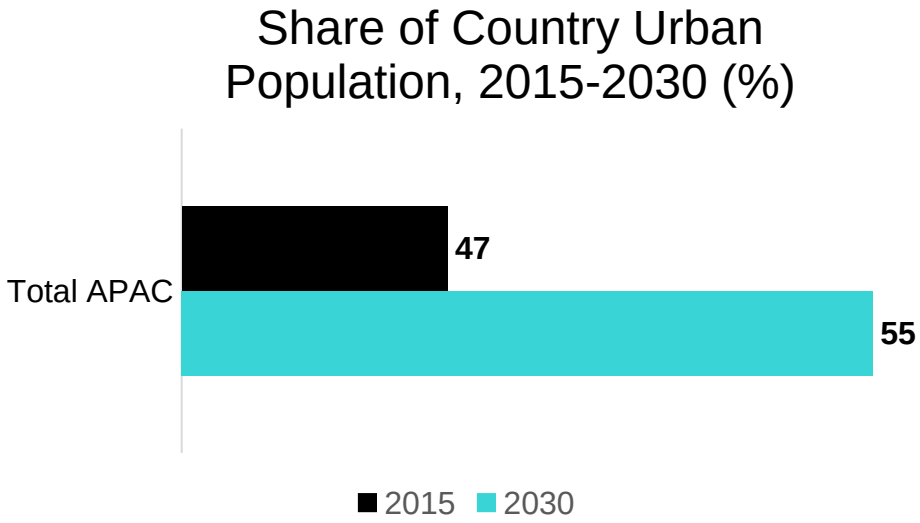
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Appendix

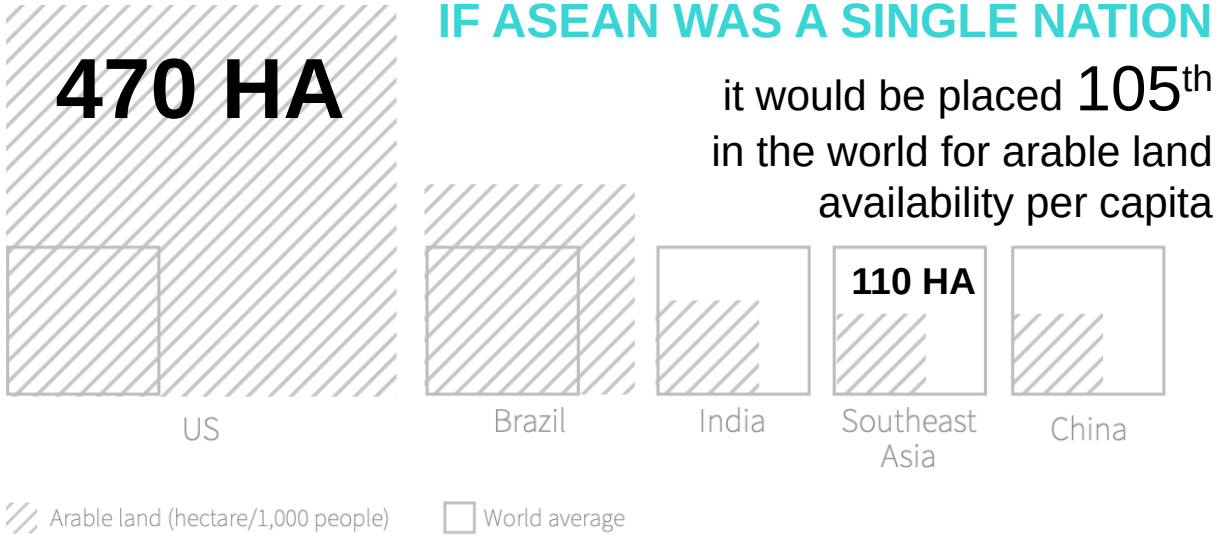
Challenges of feeding the increasing population in Urban Asia Pacific

Increasing Urban Population



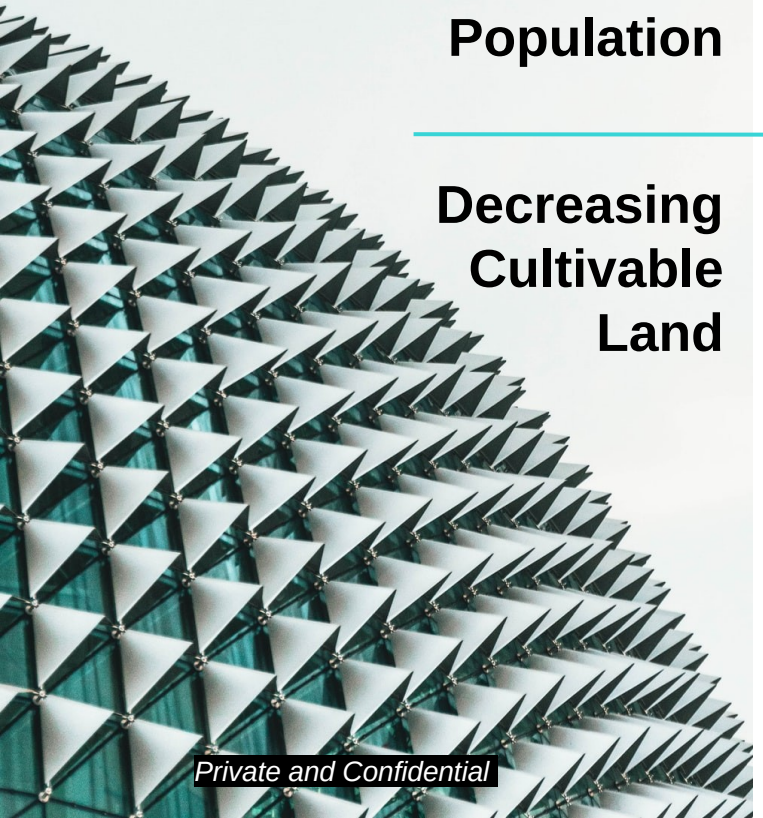
555 MILLION
people will be added to urban cities by 2030

Decreasing Cultivable Land



IF ASEAN WAS A SINGLE NATION

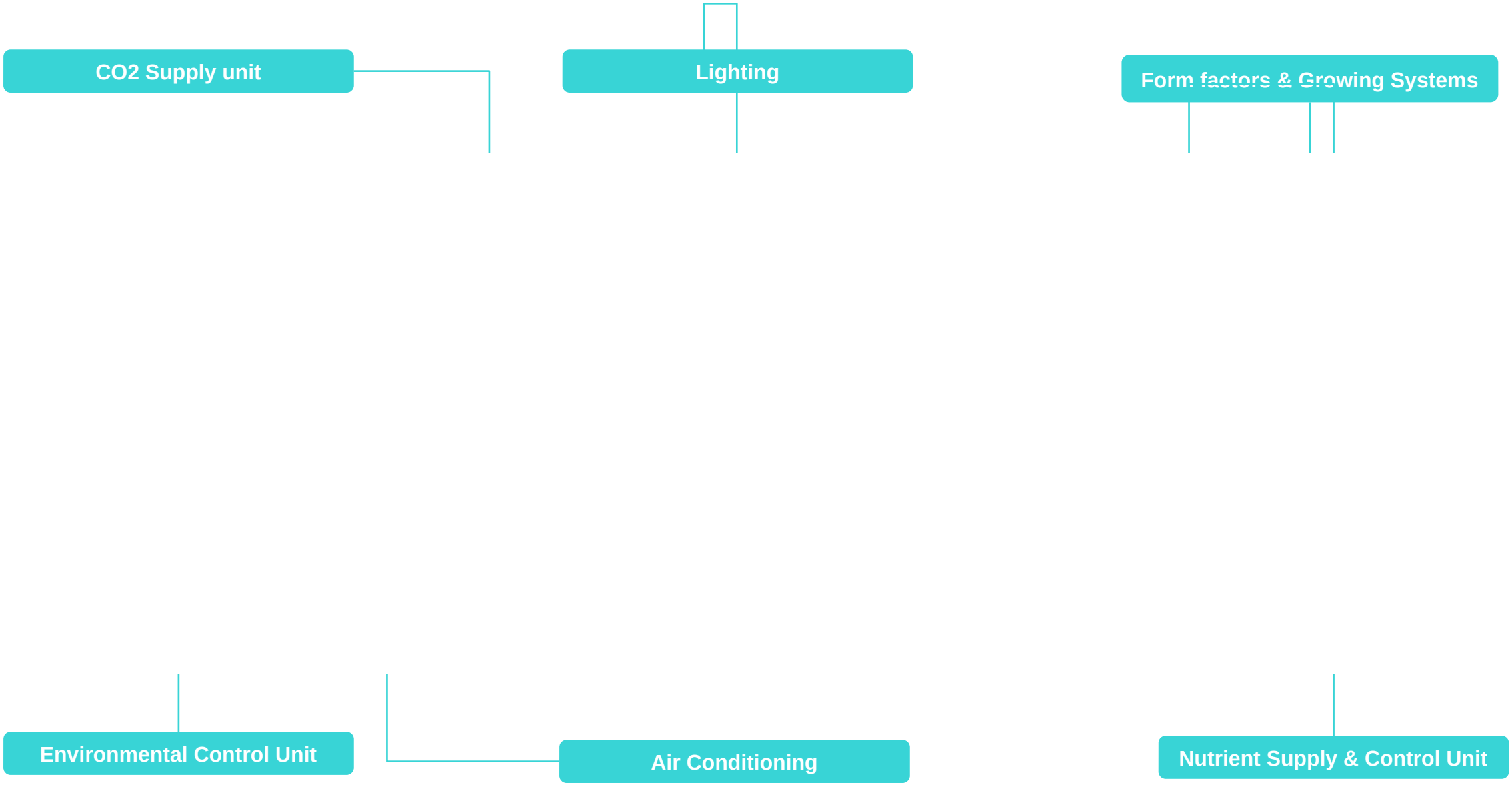
it would be placed 105th in the world for arable land availability per capita



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Source: Food and Agriculture Organization of the United Nations (2017), PWC (2020)

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[Insert Revised Business Model]

Key Partners

NUS GRIP
Farm Operato
Building Owne
HVAC & BMS Ver
Government Agen

Key Activities

ements Gathering
rm Development
tem Installation
model design,
ment, training, and
tuning

Key Resources

IP
Platform
ML Model

Value Proposition

Initial: Enable successful climate management of vertical farms by measuring key parameters and ensuring process repeatability for future harvests.

data: increase yield, o
rgy use, and reduce la
costs.

Customer Relationships

Tailored Pricing
Modern Digital Platform
Personalized Support

Channels

Direct Outreach
Equipment Suppliers
Farmer Cooperatives

Customer Segments

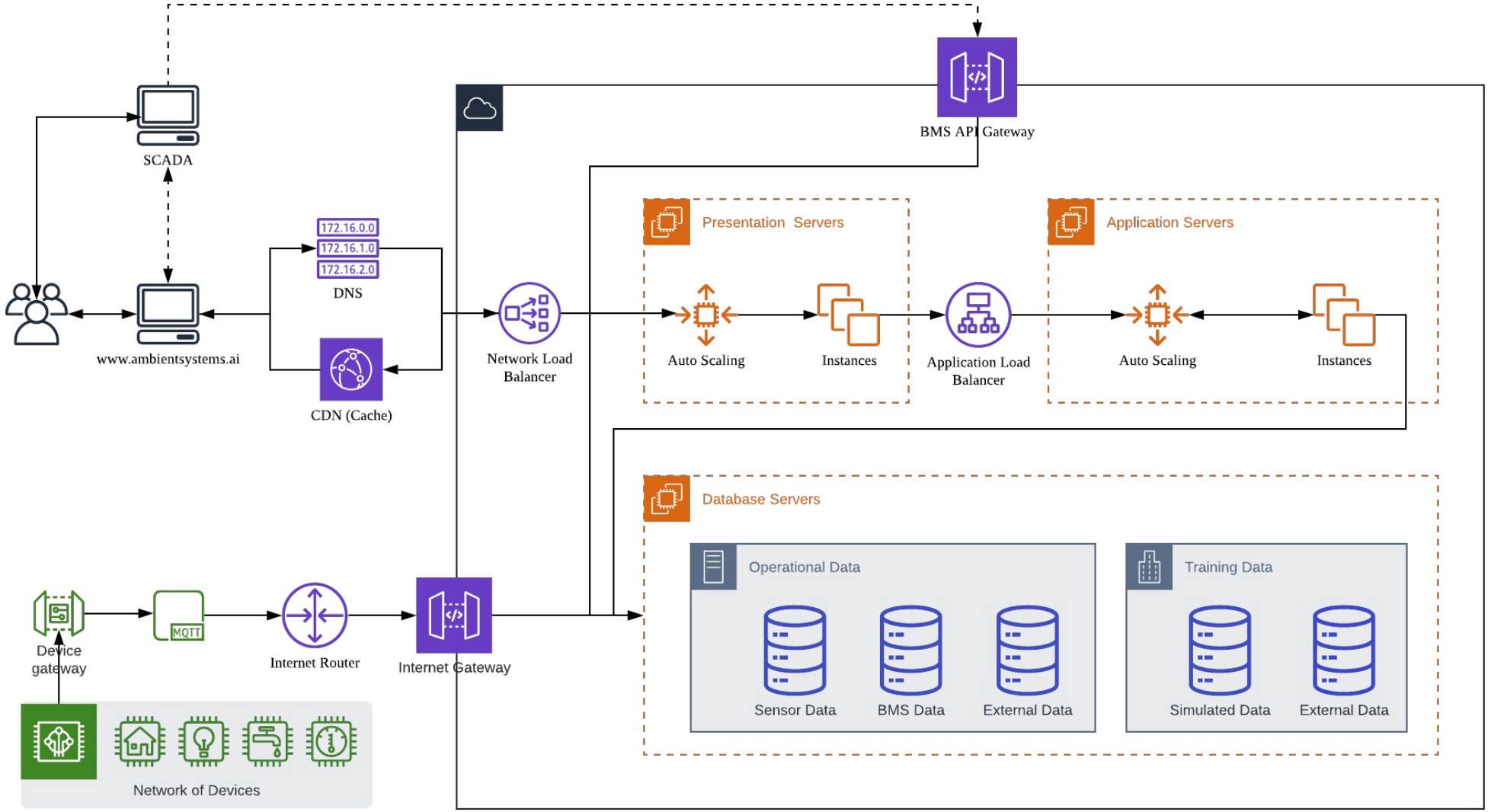
Indoor Farms

Cost Structure

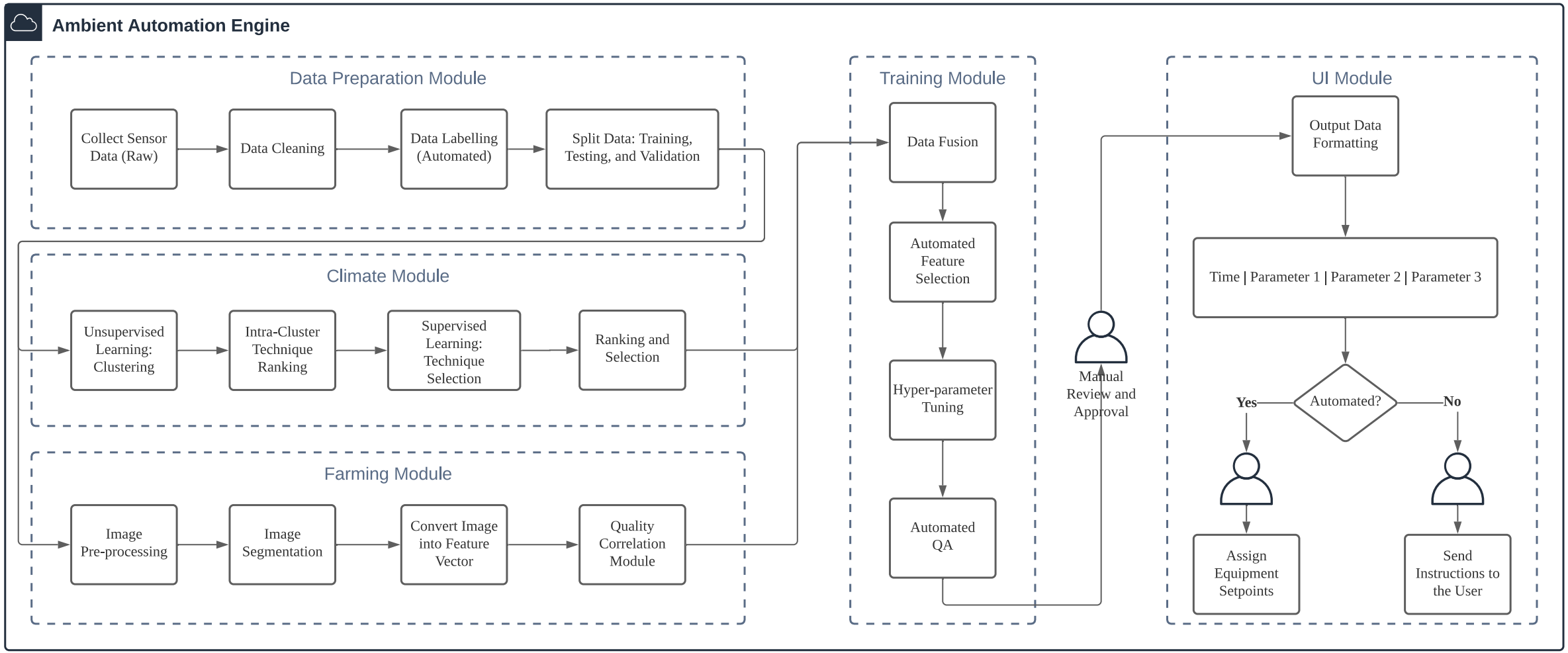
Revenue Streams

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[Insert Revised Technology Architecture]



Deep dive into the Machine Learning Optimization Model (Secret Sauce)



Patent focuses on climate optimization specific to indoor farming

We offer end-to-end climate control solutions that are guaranteed with our Pay As You Grow revenue sharing model

	AMBIENT SYSTEMS Climate At Your Service	Autogrow	75F	Johnson Controls
Segment	**Multiple**	**Indoor Farming Controller**	**Smart Buildings**	**OEM**
Proximity to Customer (Low Logistics Cost)	✓ **Yes**	✗ No	✗ No	✓ Yes
Indoor Farming Features	✓ **Yes**	✓ Yes	✗ No	✗ No
Modern Web-based UI	✓ **Yes**	✗ No	✓ Yes	✗ Some
Wireless IoT Network	✓ **Yes**	✗ No	✓ Yes	✓ Yes
Advanced Optimization Methods	✓ **Machine Learning**	✗ No	✗ Some	✗ Some
	Quotation (based on crop output)	$3000 for one unit + Installation Fees	Quotation	Quotation

Ambient aims to provide full-coverage of the Building Technology Ecosystem



2021 2022 2023 2024 2025

Software Subscription

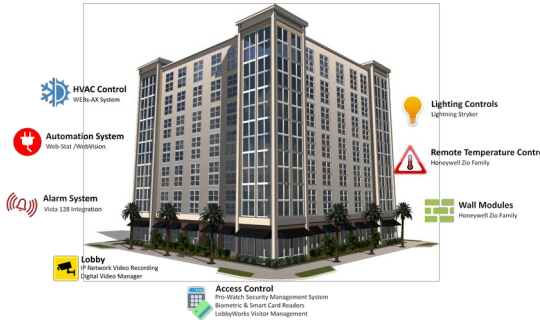
Software Subscription: Data Visualization
Software Subscription: AI-enabled Optimization
Software Subscription: Controls/Automation

OEM Market



Sensor Assembly
Sensor Integration
Sensor Manufacturing

Building Technology



Controls Assembly
Controls Integration
Controls Manufacturing (OEM)